

DAILY DRILLING REPORT - OIL-AX-22 | Report No. 45 | Date: 2022-08-13

Rig: Rig DR-02 | Field: Dikhow East (Demo Field) | Basin: Upper Assam Shelf

Depth 00:00 hrs 3,405 m MD; depth 24:00 hrs 3,478 m MD; 24-hr progress 73 m.

Hole size 8-1/2 in; last casing 9-5/8 in at 3,030 m. Formation at report depth: Kopili Shale.

Mud system: KCl-polymer WBM, mud weight 1.44 sg, ECD 1.49 sg, KCl 6.0%.

24-hr summary: Drilled Kopili Shale 3,405-3,430 m. Pulled out for bit change; tight hole at 3,432 m; back-reamed. Ran new PDC bit, bit balling at 3,455 m; tripped and changed bit. Drilled to 3,478 m.

Next 24-hr forecast: Drill ahead to planned TD 3,500 m with sweeps every stand.

OPERATIONS TIME LOG

00:00-02:00 Drilled 3,420-3,430 m, WOB 16 t, 110 RPM, 1,850 l/min, SPP 4,090 psi.

02:00-06:30 While pulling out of hole for bit change, encountered tight hole at 3,432 m with 245 kN overpull. Worked through and back-reamed 3,440-3,420 m. Blocky cavings (8% of returns) observed at shakers. Cause attributed to Kopili shale instability after 60 hours of open-hole exposure. Mitigation: back-reamed the interval, raised mud weight from 1.44 to 1.46 sg and pumped a 10 m3 high-vis sweep. NPT 4.5 hrs.

06:30-13:00 Pulled out, changed bit, ran in hole to 3,430 m. Washed last stand to bottom.

13:00-22:00 Drilled 3,430-3,455 m. At 3,455 m ROP dropped from 9 to 2 m/hr due to bit balling in sticky Kopili claystone. Pumped detergent pill with no improvement. Pulled out, found PDC cutters balled. NPT 9 hrs. Lesson: use anti-balling nozzle configuration and an ROP enhancer in the Kopili.

DRILLING PARAMETERS & REMARKS

22:00-24:00 Drilled 3,455-3,478 m. Torque began rising from 15 kN.m at 3,478 m with intermittent stick-slip; reduced WOB and increased RPM.

Mud properties at 3,478 m: MW 1.46 sg, PV 21 cP, YP 17 lb/100ft², API fluid loss 4.4 ml, KCl 6.0%, MBT 11 kg/m³.

Recommendations: maintain KCl at 7% for the remaining Kopili interval; limit open-hole exposure before trips; ream every stand through 3,420-3,440 m on the next trip.

Prepared by: Night drilling supervisor.